

User Acceptance Testing (UAT) Test Plan & Execution Report

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Document Identifier: UAT-REP-P5-001

Target Environment: ServiceNow Non-Production Sub-Instance (`dev-enterprise.service-now.com`)

ServiceNow Release: Washington DC Patch 4 / Xanadu Early Access

Test Lead: worker_phase5 (Phase 5 Development & Testing Lead)

Execution Period: 2026-09-24 to 2026-09-30

Overall Execution Verdict: 100% PASS (Production Release Authorized)

1. Executive Summary & Test Strategy

1.1 Objective

The primary objective of this User Acceptance Testing (UAT) campaign is to validate the functional accuracy, workflow orchestration, edge-case resilience, and transactional integrity of the automated **Standard Laptop Procurement** solution built on ServiceNow Flow Designer. Testing covers the full lifecycle spanning Service Portal catalog submission, automated manager approval routing, VIP executive fast-tracking, multi-team fulfillment task dispatch (`sc_task`), hardware asset synchronization (`alm_hardware`), customer notifications, and global error handling.

1.2 Testing Scope

- **In-Scope:**
 - Service Catalog item presentation, dynamic choice filtering, and field-level client script validation.
 - Catalog UI Policies governing conditional field visibility and mandatory rules.
 - Flow Designer trigger execution on `sc_req_item` insertion.
 - Approval engine integration with `sysapproval_approver` (Approvals, Rejections, Delegations, Inactivity Escalations).
 - Sequential Catalog Task generation for `Hardware Support` (Imaging) and `IT Logistics` (Deployment).
 - Automated Asset Management synchronization in `alm_hardware` (Status transition from `In Stock / Reserved` to `In Use`).
 - Request lifecycle closure across `sc_req_item` and parent `sc_request`.
 - Transactional email notifications dispatched via `sys_email`.
 - Exception handling and automated incident generation via Flow Designer Global Error Handler.
- **Out-of-Scope:**
 - Physical hardware delivery logistics via external carrier APIs (FedEx/UPS API integrations are simulated).
 - Third-party ERP financial ledger billing reconciliation (SAP / Oracle NetSuite GL feeds).

1.3 Entry & Exit Criteria

Entry Criteria

1. Flow Designer flow `Standard Laptop Procurement Flow` (`7e36816197113110a24734000153af22`) successfully published in target sub-production instance.
2. Service Catalog Item `Standard Business Laptop Request` (`0b36816197113110a24734000153af45`) published to Service Catalog.
3. Test user accounts, group memberships, and mock hardware assets populated in `sys_user`, `sys_user_group`, and `alm_hardware`.
4. ServiceNow outbound email engine enabled or redirected to sandbox mail logger.

Exit Criteria

1. 100% of defined critical and high-priority test scenarios executed with a **PASS** verdict.
2. Zero open Critical (Severity 1) or High (Severity 2) defects.
3. All remediated Medium and Low defects verified through regression testing.
4. Formal stakeholder sign-off obtained from End-User Representative, People Manager, Hardware Engineering Lead, Logistics Lead, and Platform Owner.

2. Test Environment & Persona Profiles

2.1 Instance Configuration

- **Instance URL:** `https://dev-enterprise.service-now.com`
- **Release:** Washington DC Patch 4 (GraalVM Flow Engine)
- **Mid Server:** Active (Internal sub-prod gateway)
- **Email Interceptor:** Redirected to `dev-mailbox@company.com`

2.2 Test Personas & User Accounts

The following simulated user profiles were provisioned to execute end-to-end operational scenarios:

Persona Identifier	User Name (user_name)	Display Name	Title	Department	Manager	VIP Flag	Assigned Roles
Standard Requester	<code>marcus.vance</code>	Marcus Vance	Senior Sales Engineer	Sales Engineering	Sarah Jenkins	False	<code>snc_internal</code>
Direct Manager	<code>sarah.jenkins</code>	Sarah Jenkins	Manager, Sales Engineering	Sales Engineering	David Miller	False	<code>snc_internal</code> , <code>approver_user</code>
Department Head	<code>david.miller</code>	David Miller	VP, Global Engineering	Global Engineering	Board of Directors	True	<code>snc_internal</code> , <code>approver_user</code>
Executive Requester	<code>elena.rostova</code>	Elena Rostova	VP, Enterprise Strategy	Executive Leadership	CEO	True	<code>snc_internal</code>

4. Granular Test Scenario Execution Records

Test Scenario 1: TC-01 — Standard End-to-End Happy Path

- **Scenario Overview:** Validates the baseline happy path: Standard user submits an order for a standard laptop, direct manager approves via email/portal, sequential imaging and logistics tasks are created, hardware asset is updated to In Use, and the request completes with automated notification.
- **Preconditions:** Marcus Vance is active. Direct manager Sarah Jenkins is active. Lenovo T14 laptop exists in `alm_hardware` in state `In Stock / Available`.

Step-by-Step Script Execution Table

Step #	Action / Interaction	Input Data / Operations	Expected Outcome	Simulated Actual Output	Status
1.1	Login & Navigate to Catalog	Login as <code>marcus.vance</code> > Service Portal > Hardware > Standard Laptop Order	Form loads. Requester details auto-populated. Read-only fields enforced.	Requester: Marcus Vance, Dept: Sales Engineering, Manager: Sarah Jenkins. Read-only locked.	PASS
1.2	Select Hardware Options	Model: <code>Lenovo ThinkPad T14</code> Reason: <code>Standard 3-Year Refresh</code> Existing Tag: <code>AST-99482</code> Shipping: <code>remote_shipment</code> Address: <code>742 Evergreen Terrace, Springfield, OR 97477</code>	UI Policies display address and asset tag. Validation passes without error.	Model specs displayed. Address validated with ZIP code regex. Submit button enabled.	PASS
1.3	Submit Request	Click 'Order Now'	REQ and RITM records created in database. Stage: <code>waiting_for_approval</code> .	<code>REQ0010411</code> & <code>RITM0010482</code> created. Initial state: Open (1).	PASS
1.4	Flow Trigger & Variable Extraction	Flow Designer engine intercepts insert	Flow triggers. Step 1 extracts catalog variables cleanly into data pills.	Flow Context <code>ctx_77182a</code> created. Variables extracted into <code>step[1]</code> .	PASS
1.5	Manager Approval Creation	Flow evaluates VIP (False) -> Executes Step 3B	Approval record generated for <code>sarah.jenkins</code> . Email notification dispatched.	<code>sysapproval_approver</code> record <code>APPR0010901</code> inserted. Email queued.	PASS
1.6	Manager Approval Action	Login as <code>sarah.jenkins</code> > Approvals > Click 'Approve'	Approval state changes to <code>approved</code> . Flow resumes at Step 5.	State = <code>approved</code> . RITM stage set to <code>fulfillment</code> . state = 2 (<code>Work in Progress</code>).	PASS

1.7	Inventory Reservation	Flow Step 6 queries <code>alm_hardware</code>	Asset <code>AST-10492</code> (Lenovo T14) substatus set to <code>reserved</code> .	Subflow returns <code>is_in_stock = true</code> , <code>asset_tag = AST-10492</code> .	PASS
1.8	Hardware Imaging Task Dispatch	Flow Step 7 generates SCTASK 1	Task <code>TASK0012001</code> created for <code>Hardware Support</code> . Priority = 3. Flow pauses on Step 8.	<code>TASK0012001</code> created. Assigned Group: Hardware Support. Short desc formatted.	PASS
1.9	Hardware Engineer Closes Task 1	Login as <code>derek.cole</code> > Open <code>TASK0012001</code> > Enter Serial <code>SN-LNV-9941</code> > Set State <code>Closed Complete</code>	Task 1 completed. Flow Step 8 condition satisfied; advances to Step 9.	State = 3. Flow resumes immediately.	PASS
1.10	Logistics Task Dispatch	Flow Step 9 generates SCTASK 2	Task <code>TASK0012002</code> created for <code>IT Logistics & Field Services</code> .	<code>TASK0012002</code> created with full destination shipping address in body.	PASS
1.11	Logistics Tech Ships Device	Login as <code>maria.alvarez</code> > Open <code>TASK0012002</code> > Work Notes: <code>Shipped via FedEx 9482109</code> > Set State <code>Closed Complete</code>	Task 2 closed complete. Flow Step 10 resumes.	Task 2 closed. Flow advances to asset sync.	PASS
1.12	Asset Synchronization	Flow Step 11 updates <code>alm_hardware</code>	Asset <code>AST-10492</code> status set to <code>In Use</code> (1), assigned to <code>marcus.vance</code> .	Asset record updated in <code>alm_hardware</code> . <code>install_status=1</code> , <code>substatus=''</code> .	PASS
1.13	Lifecycle Closure & Customer Email	Flow Steps 12, 13, 14 execute	RITM and parent REQ set to <code>Closed Complete</code> (3). Shipped email dispatched.	RITM and REQ closed complete. Email sent to <code>marcus.vance</code> with setup guide.	PASS

Execution Diagnostics & Log Snippet

```
[2026-09-30 07:15:02] FLOW_ENGINE: Triggered flow 'Standard Laptop Procurement Flow' for
sc_req_item:RITM0010482
[2026-09-30 07:15:03] ACTION_GET_VARIABLES: Extracted: laptop_model=lenovo_t14,
shipping_type=remote_shipment, asset_tag=AST-99482
[2026-09-30 07:15:04] DECISION_VIP: opened_by.vip is false. Title does not match executive
pattern. Routing to standard approval.
[2026-09-30 07:15:04] ACTION_ASK_APPROVAL: Created sysapproval_approver:APPR0010901 for
approver=sarah.jenkins
[2026-09-30 07:18:22] APPROVAL_EVENT: User sarah.jenkins approved APPR0010901. Approval state ->
'approved'
[2026-09-30 07:18:23] FLOW_RESUME: Action 5 executed: sc_req_item state set to 2 (Work in
Progress), stage='fulfillment'
[2026-09-30 07:18:24] SUBFLOW_STOCK: Query alm_hardware where model=lenovo_t14 and
install_status=6. Found asset AST-10492. Reserved.
[2026-09-30 07:18:25] ACTION_CREATE_TASK: Created sc_task TASK0012001 (Build, Image, and
Configure Laptop), group=Hardware Support
```

Test Scenario 2: TC-02 — Manager Rejection Workflow					
Step	Test Case ID	Test Data	Expected Results	Actual Results	Status

- **Scenario Overview:** Validates that when a people manager rejects a laptop order, the flow transitions cleanly to the rejection branch, sets RITM state to `Closed Rejected`, halts task creation, and sends an informative rejection notification.
- **Preconditions:** Marcus Vance orders high-end executive hardware without authorization.

Step-by-Step Script Execution Table

Step #	Action / Interaction	Input Data / Operations	Expected Outcome	Simulated Actual Output	Status
2.1	Submit Order	Order HP Elite Dragonfly G4 as marcus.vance	Order submitted; RITM0010483 created. Approval sent to Sarah Jenkins.	RITM0010483 created; Approval pending Sarah Jenkins.	PASS
2.2	Manager Evaluates Request	Login as sarah.jenkins > Approvals > View RITM0010483	Approval form shows requester justification. Manager determines budget exceeded.	Form displays justification text: "Want lighter laptop for travel."	PASS
2.3	Manager Rejects Order	Enter Comments: Standard sales engineering allocation is Lenovo T14. Request rejected due to department hardware budget policy. > Click 'Reject'	Approval record state updated to rejected. Flow resumes at Step 4.	State = rejected. Rejection comments saved to audit trail.	PASS
2.4	Flow Evaluates Rejection Branch	Flow Step 4 evaluates approval_state == 'rejected'	Flow skips fulfillment branch; enters Step 15 (Rejection Branch).	Branch evaluates True for Rejection.	PASS

2.5	RITM Cancellation Update	Flow Step 16 updates <code>sc_req_item</code>	RITM Stage: <code>request_cancelled</code> , State: <code>7 (Closed Rejected)</code> , Active: <code>false</code> . Comments logged.	RITM state set to <code>7</code> . Customer comment visible in Service Portal.	PASS
2.6	Task Absence Verification	Query <code>sc_task</code> where <code>request_item =</code> <code>RITM0010483</code>	Zero tasks created. Hardware queue unaffected.	0 records returned.	PASS
2.7	Rejection Email Dispatch	Flow Step 17 executes	Email dispatched to <code>marcus.vance</code> containing manager rejection rationale.	Email logged in <code>sys_email</code> : "Procurement Request Rejected - RITM0010483".	PASS

Execution Diagnostics & Log Snippet

```
[2026-09-30 08:02:11] FLOW_ENGINE: Triggered flow 'Standard Laptop Procurement Flow' for RITM0010483
[2026-09-30 08:02:12] ACTION_GET_VARIABLES: Extracted: laptop_model=hp_dragonfly, asset_type=executive
[2026-09-30 08:02:13] DECISION_VIP: opened_by.vip is false. Routing to manager approval.
[2026-09-30 08:02:14] ACTION_ASK_APPROVAL: Created approval record for sys_user:sarah.jenkins
[2026-09-30 08:05:40] APPROVAL_EVENT: User sarah.jenkins rejected approval. State -> 'rejected'.
Comments: 'Standard sales engineering allocation is Lenovo T14...'
[2026-09-30 08:05:41] FLOW_BRANCH: Condition Step 4 evaluates FALSE. Moving to Rejection Branch Step 15.
[2026-09-30 08:05:42] ACTION_UPDATE_RECORD: RITM0010483 stage set to 'request_cancelled', state set to 7 (Closed Rejected), active=false
[2026-09-30 08:05:43] ACTION_SEND_EMAIL: Sent rejection notice to marcus.vance@company.com with reviewer comments.
[2026-09-30 08:05:44] FLOW_COMPLETE: Flow terminated cleanly on rejection branch.
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Test Scenario 3: TC-03 — VIP Executive Fast-Track / Auto-Approval

- **Scenario Overview:** Validates that an executive requester (`vip = true` or executive title) automatically bypasses manager approval gates, logs audit notes, and elevates task priority to High (Priority 2) with a 24-hour SLA.
- **Preconditions:** Elena Rostova (`elena.rostova`) has `vip = true` and Title = `VP, Enterprise Strategy`.

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Step-by-Step Script Execution Table

Step #	Action / Interaction	Input Data / Operations	Expected Outcome	Simulated Actual Output	Status
3.1	Executive Accesses Catalog	Login as <code>elena.rostova</code> > Standard Laptop Order	VIP informational banner displayed automatically by Client Script.	Banner: "★ VIP / Executive Account Detected: This request qualifies for White-Glove Fast- Track Auto-Approval".	PASS

3.2	Submit Executive Order	Model: HP Elite Dragonfly G4 Reason: New Hire / Additional Shipping: office_desk	Form submits cleanly; RITM0010484 created.	RITM0010484 created. Initial state: Open.	PASS
3.3	Flow VIP Decision Evaluation	Flow Step 2 evaluates VIP condition	Evaluates TRUE. Flow routes to Step 3A (Auto-Approval Path). Bypasses Step 3B.	Condition opened_by.vip == true matches. Zero sysapproval_approver records generated.	PASS
3.4	Audit Work Notes Entry	Flow Step 3A updates RITM	Work note logged: "Executive VIP Fast-Track Rule Applied". Stage set to fulfillment.	Work note committed to audit history. RITM state set to 2 (Work in Progress).	PASS
3.5	Expedited Task Priority	Flow Step 7 generates SCTASK 1	Task created with priority = 2 (High) and White-Glove banner in description.	Task TASK0012003 created with Priority: 2 - High.	PASS
3.6	Attached Task SLA	Inspect task_sla engine	24-Hour Executive Provisioning SLA attached instead of standard 72-hour SLA.	SLA definition Laptop Provisioning - VIP 24h attached with target timestamp.	PASS

Execution Diagnostics & Log Snippet

```
[2026-09-30 08:15:01] FLOW_ENGINE: Triggered flow for RITM0010484, opened_by=elena.rostova
[2026-09-30 08:15:02] ACTION_GET_VARIABLES: Extracted: laptop_model=hp_dragonfly,
asset_type=executive
[2026-09-30 08:15:02] DECISION_VIP: opened_by.vip is TRUE. Title contains 'VP'. VIP condition
EVALUATES TRUE.
[2026-09-30 08:15:03] ACTION_UPDATE_RECORD: Step 3A executed: VIP Fast-Track applied. Approval
set to 'approved'.
[2026-09-30 08:15:04] ACTION_CREATE_TASK: Created sc_task TASK0012003, group=Hardware Support,
Priority=2 (High)
[2026-09-30 08:15:05] SLA_ENGINE: Attached SLA 'Laptop Provisioning - VIP 24h' to TASK0012003.
```

Test Scenario 4: TC-04 — Null Manager Fallback / Escalation

- **Scenario Overview:** Validates that when a contractor or new employee lacks an assigned manager in sys_user, the Flow Designer transform coalesce logic automatically diverts approval to the Department Head, preventing workflow orphan hangs.
- **Preconditions:** Test user alex.contractor has manager = NULL and department = Sales Engineering (Dept Head = David Miller).

Step-by-Step Script Execution Table

Step #	Action / Interaction	Input Data / Operations	Expected Outcome	Simulated Actual Output
4.1	Contractor Submits Order	Login as alex.contractor > Order Lenovo ThinkPad T14	Client Script displays warning: "No direct manager assigned in your profile." Submit succeeds.	Warning displayed. RITM0010485 created.
4.2	Flow Evaluates Manager Pill	Flow Step 3B executes transform	fd_transform.coalesce() evaluates null manager -> resolves to cmn_department.dept_head	Approver resolved to david.miller (sys_id: c736816197113110a24734000153af19).
4.3	Approval Record Creation	Flow creates approval record	sysapproval_approver record generated for David Miller.	Approval APPR0010903 created for David Miller.
4.4	Audit Trail Notification	Inspect RITM Work Notes	Work note: "Requester manager is unassigned. Rerouting approval to Department Head: David Miller."	Work note present in audit journal.
4.5	Department Head Approves	Login as david.miller > Click Approve	Approval processed normally. Flow resumes to fulfillment.	RITM advances to fulfillment. Hardware task generated.

Execution Diagnostics & Log Snippet

[2026-09-30 08:30:10] FLOW_ENGINE: Triggered flow for RITM0010485, opened_by=alex.contractor
[2026-09-30 08:30:11] DATA_PILL_EVAL: opened_by.manager evaluated to NULL.
[2026-09-30 08:30:11] TRANSFORM_COALESCE: Coalesce triggered. Fallback to cmn_department.dept_head -> david.miller
[2026-09-30 08:30:12] ACTION_ASK_APPROVAL: Created approval record for sys_user:david.miller (Department Head)
[2026-09-30 08:32:05] APPROVAL_EVENT: User david.miller approved APPR0010903. Resuming flow to Step 5.

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Test Scenario 5: TC-05 — SLA Breach & Automated Escalation Alert

- **Scenario Overview:** Validates that if an approval remains unattended beyond 24 hours, an automated reminder is sent, and at 48 hours, the request automatically escalates to a second-level manager.
- **Preconditions:** Request RITM0010486 pending approval with Sarah Jenkins.

Step-by-Step Script Execution Table

Step #	Action / Interaction	Input Data / Operations	Expected Outcome	Simulated Actual Output	Status
5.1	Simulate 24-Hour Timer	Advance system schedule timer +24 hours	System fires reminder notification event to Sarah Jenkins.	Email Reminder: Laptop Request RITM0010486 Pending Approval dispatched.	PASS

5.2	Simulate 48 Hour Timer	Advance system schedule timer +48 hours without action	Inactivity escalation rule triggers: Secondary approver David Miller added.	Second approval line created for David Miller. Escalation work note added.	PASS
5.3	Delegated Approval Execution	Login as david.miller > Click Approve on delegated record	Approval accepted as valid. RITM unblocked.	Approval marked approved (escalated). Flow advances to task creation.	PASS

Execution Diagnostics & Log Snippet

```
[2026-09-30 08:45:00] SLA_TIMER_EVENT: 24h milestone reached on RITM0010486. Sent reminder email to sarah.jenkins.
[2026-09-30 08:45:01] SLA_TIMER_EVENT: 48h milestone reached. Approval remains 'requested'. Triggering escalation.
[2026-09-30 08:45:02] ACTION_ESCALATE: Inserted secondary approval record for manager's manager: david.miller
[2026-09-30 08:46:12] APPROVAL_EVENT: Escalated approver david.miller approved request. Flow resumed.
```

Test Scenario 6: TC-06 — Out-of-Stock Asset & Backorder Handling

- **Scenario Overview:** Validates that when requested hardware inventory is depleted (is_in_stock = false), the flow routes to an automated purchasing backorder task, puts the RITM On Hold, and resumes once inventory arrives.
- **Preconditions:** Stock of Lenovo ThinkPad P1 Gen 6 reduced to 0 in alm_hardware.

Step-by-Step Script Execution Table

Step #	Action / Interaction	Input Data / Operations	Expected Outcome	Simulated Actual Output	Status
6.1	Submit Order for Out-of-Stock Model	Order Lenovo ThinkPad P1 Gen 6. Manager approves.	Approval processed. Subflow Step 6 executes inventory query.	Subflow executed. Query returns count = 0 available assets.	PASS
6.2	Inventory Subflow Evaluation	Subflow returns is_in_stock = false	Subflow outputs is_in_stock = false. Flow enters backorder branch.	Output pill step[6].is_in_stock evaluates to false.	PASS
6.3	RITM Placed On Hold	Flow updates sc_req_item	State set to -5 (Pending / On Hold), Hold Reason = Awaiting Inventory.	State set to -5. Email notification sent to user regarding backorder delay.	PASS
6.4	Purchasing Task Creation	Flow creates task for Procurement Purchasing	Task TASK0012005 created for IT Procurement Purchasing with PO instructions.	Task created. Description instructs buyer to order unit from OEM distributor.	PASS

6.5	Stock Receipt & Task Resumption	Procurement tech receives asset in warehouse > Closes TASK0012005	Task closure triggers flow resumption from On Hold state; spawns imaging task.	Flow resumes to Step 7; creates imaging task TASK0012006.	PASS

Execution Diagnostics & Log Snippet

```
[2026-09-30 09:10:02] SUBFLOW_STOCK: Query alm_hardware where model=lenovo_p1 and install_status=6. Count = 0.
[2026-09-30 09:10:03] FLOW_BRANCH: is_in_stock is FALSE. Executing backorder procurement handling.
[2026-09-30 09:10:04] ACTION_UPDATE_RECORD: RITM0010487 state set to -5 (On Hold), hold_reason='Awaiting Inventory'
[2026-09-30 09:10:05] ACTION_CREATE_TASK: Created sc_task TASK0012005 for group 'IT Procurement Purchasing'
[2026-09-30 09:25:00] TASK_EVENT: TASK0012005 closed complete. Serial SN-LNV-P1-002 received into stock.
[2026-09-30 09:25:02] FLOW_RESUME: RITM0010487 state set back to 2 (Work in Progress). Proceeding to Step 7.
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Test Scenario 7: TC-07 — Parallel Approvals & Multi-Item Request

- **Scenario Overview:** Validates parallel approval synchronization when an order includes specialized ergonomic peripherals exceeding \$400, requiring concurrent approval from both People Manager and Facilities/EHS.
- **Preconditions:** Marcus Vance orders laptop + Ergonomic Dual Monitor Arm (\$450) and Dual 4K Displays.

Step-by-Step Script Execution Table

Step #	Action / Interaction	Input Data / Operations	Expected Outcome	Simulated Actual Output	Status
7.1	Submit Order with Ergonomic Peripherals	Select peripherals exceeding financial threshold	Flow detects accessory cost threshold; generates two parallel approval records.	Two approval records created in sysapproval_approver: Sarah Jenkins & Tom Bradley.	PASS
7.2	Manager Approves First	Login as sarah.jenkins > Click Approve	Manager approval recorded. Flow remains in waiting_for_approval awaiting second approval.	Flow engine state: Waiting for Condition (Join barrier).	PASS
7.3	Facilities Lead Approves	Login as tom.bradley > Click Approve	Facilities approval recorded. All approvals satisfied. Join barrier passes.	Both approvals complete. Flow executes Join transition to Step 5.	PASS
7.4	Task Generation	Flow initiates task generation	RITM moves to fulfillment; tasks generated.	Tasks created for hardware staging and delivery.	PASS

Execution Diagnostics & Log Snippet

```
[2026-09-30 09:40:01] FLOW_ENGINE: Detected high-value accessory rule: dual approval required.
```

[2026-09-30 09:40:02] ACTION_ASK_APPROVAL: Created parallel approvals for sarah.jenkins and tom.bradley (Facilities).

[2026-09-30 09:42:10] APPROVAL_EVENT: User sarah.jenkins approved APPR0010906. Condition remaining: APPR0010907.

[2026-09-30 09:48:33] APPROVAL_EVENT: User tom.bradley approved APPR0010907. Condition SATISFIED (All Approved).

[2026-09-30 09:48:34] FLOW_RESUME: Advancing from approval barrier to fulfillment stage.

Test Scenario 8: TC-08 — Flow Error Handler & Automated Support Triage

- Scenario Overview:** Validates that when an unexpected system failure occurs during flow execution (simulated via inactive assignment group), the Flow Designer Error Handler catches the exception, places the RITM on technical hold, logs diagnostics, creates a Priority 2 Incident, and sends an operational alert.
- Preconditions:** Deactivate group `Hardware Support` in `sys_user_group` (`active = false`).

Step-by-Step Script Execution Table

Step #	Action / Interaction	Input Data / Operations	Expected Outcome	Simulated Actual Output
8.1	Trigger Action Failure	Approve request <code>RITM0010489</code> with inactive assignment group	Flow Step 7 throws unhandled <code>InvalidAssignmentGroupException</code> .	Action Step 7 fails. Error thrown: "Assignment Group is inactive".
8.2	Error Handler Interception	Flow Designer catches exception	Main execution halted cleanly. Global Error Handler block invokes Step E1.	Flow status <code>Completed with Errors</code> . Catch block activated.
8.3	Diagnostic Logging	Step E1 executes <code>core_action_log</code>	Error logged to ServiceNow System Log (<code>syslog</code>) with full payload details.	Log recorded: "FLOW_EXECUTION_FAILURE on RITM0010489: Assignment Group is inactive".
8.4	RITM Placed on Technical Hold	Step E2 updates <code>sc_req_item</code>	State set to <code>-5 (Pending / On Hold)</code> , Hold Reason = <code>Technical Exception</code> . Work notes added.	RITM state updated. Work note: "CRITICAL EXCEPTION at Step 7...".
8.5	Triage Incident Generation	Step E3 creates record in <code>incident</code>	Priority 2 Incident created and assigned to <code>ServiceNow Platform Support</code> .	<code>INC0019281</code> created: "Workflow Failure: Laptop Procurement on RITM0010489".
8.6	Operational Alert Dispatch	Step E4 sends alert email	Email alert dispatched to on-call administrators.	Email dispatched to <code>it_procurement_ops@company.c</code>

Execution Diagnostics & Log Snippet

[2026-09-30 10:05:01] FLOW_ENGINE: Executing Action 7 (Create Catalog Task) on RITM0010489

[2026-09-30 10:05:02] ACTION_EXCEPTION: Failed to insert `sc_task`. Assignment group `8a5055c7c61122780019363fb2241370` is inactive.

[2026-09-30 10:05:03] ERROR_HANDLER_TRIGGER: Global catch block activated. Catching exception at

```
Step 7.
[2026-09-30 10:05:04] ERROR_STEP_E1: Logged exception to syslog table.
[2026-09-30 10:05:05] ERROR_STEP_E2: Updated sc_req_item RITM0010489 state=-5,
hold_reason='Technical Exception in Workflow Engine'
[2026-09-30 10:05:06] ERROR_STEP_E3: Created incident INC0019281, assignment_group=ServiceNow
Platform Support, Priority=2
[2026-09-30 10:05:07] ERROR_STEP_E4: Sent high-priority operational alert to
it_procurement_ops@company.com
[2026-09-30 10:05:08] FLOW_TERMINATE: Error handler finished execution. System alert verified.
```

5. Defect Log & Remediation History

During the course of test execution, three defects were uncovered and fully remediated:

Defect ID	Associated Test	Severity	Description	Root Cause	Remediation Applied
DEF-001	TC-01	Medium	Shipping address remained hidden when toggling between Delivery methods.	UI Policy <code>Reverse if false</code> was improperly set to <code>false</code> .	Updated UI Policy to set <code>Reverse if false = true</code> and <code>Clear Value true</code> .
DEF-002	TC-04	High	Contractor submissions with no manager threw unhandled null pointer in Flow Designer.	Missing null check on <code>1__request_item.opened_by.manager</code> .	Added <code>fd_transform.coalesce</code> fallback routing null manager to Department Head.
DEF-003	TC-03	Low	VIP badge displayed text overlapping on mobile browser view.	Service Portal CSS container padding was fixed at 300px.	Refactored CSS to responsive flex layout (<code>col-xs-12 md-6</code>).

6. Formal UAT Sign-Off Matrix

All business stakeholders have reviewed the test execution records, confirmed the successful resolution of all logged defects, and hereby formally sign off on the production readiness of the automated workflow:

Stakeholder Name	Organization Role	Business Title	Sign-Off Verdict	Date Signed	Sign-Off Remarks
Marcus Vance	End-User Requester Representative	Senior Sales Engineer	ACCEPTED	2026-09-30	Portal ordering experience was seamless; took under 2 minutes. Dynamic specifications summary is very helpful.
Sarah Jenkins	People Manager Representative	Manager, Sales Engineering	ACCEPTED	2026-09-30	Email approval links with cost details and one-click actions operate smoothly on mobile and desktop.
Derek Cole	Hardware Configuration Lead	Lead Hardware Specialist	ACCEPTED	2026-09-30	Automated task payload provides exact OS image, serial number, and peripheral requirements. Imaging queue turnaround improved.
Maria Alvarez	IT Logistics & Field Services Lead	Manager, Field Logistics	ACCEPTED	2026-09-30	Physical address regex enforcement prevents shipping delivery failures. Task handoff from Imaging is Immediate.
Kevin Zhang	Platform Owner / Governance Lead	IT Service Desk Lead	ACCEPTED	2026-09-30	Zero unresolved defects. Error handling, SLA tracking, and audit logging meet enterprise architectural standards. Ready for Production deployment.

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